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BSIT-22D

Data Communication and Networking 2
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1. What are Intelligent Connected Vehicles (ICVs)? 5 points

   Ans: ICVs are smart cars that use sensors, AI, and the internet to talk to other cars and roads. They help you drive safer and easier by avoiding crashes and traffic jams. Some can drive themselves a little, others a lot. They're the future of driving, especially in smart cities.

2. What is the first phase in the development of ICVs? 5 points

   Ans: It's called the Basic Assistance Phase. It introduces foundational ADAS features. Focuses on driver alerts and limited automation (e.g., lane departure warnings). Relies on onboard sensors like cameras and radar. No vehicle-to-everything (V2X) communication yet. The driver remains fully responsible for control.

3. What driver-assistance features are included in the Basic Assistance Phase? 5 points

   Ans: These are the following:

   Lane Departure Warning (LDW)

   Forward Collision Warning (FCW)

   Automatic Emergency Braking (AEB)

   Adaptive Cruise Control (ACC)

   Blind Spot Detection (BSD)

4. How does the Connected Vehicle Phase improve driving experience? 5 points

   Ans: Enables real-time traffic and hazard alerts via V2V/V2I. It optimizes routing using cloud-based traffic data. It enhances safety with cooperative awareness (e.g., red-light warnings). It also reduces congestion through coordinated vehicle flow. And supports over-the-air (OTA) updates for feature improvements.

5. What is the role of GPS in Intelligent Connected Vehicles? 5 points

   Ans: Provides precise location data for navigation and mapping. It enables geofencing and route optimization. It supports V2I communication (e.g., traffic signal timing). It will assist in autonomous path planning and lane-level positioning. It will integrate with other sensors for sensor fusion accuracy.

6. What is the difference between the Semi-Autonomous Phase and the Highly Autonomous Phase? 5 points

   Ans:

Semi-Autonomous Phase	Highly Autonomous Phase
Driver must monitor and intervene (e.g., Tesla Autopilot)	The vehicle handles most scenarios; driver optional in specific conditions.
It still requires human oversight; highly autonomous reduces it significantly.	It uses more advanced AI and sensor fusion.

7. Why is driver supervision still necessary in semi-autonomous vehicles? 5 points

   Ans: Because systems can't handle all edge cases or unpredictable environments. And legal liability still rests with the human driver. Sensors/AI may fail or misinterpret complex situations. It ensures safety during system transitions or failures. Regulatory standards require human readiness to take over.

8. What technologies are commonly used in Intelligent Connected Vehicles? 5 points

   Ans:

   - LiDAR, radar, and cameras for perception.
   - AI/ML for decision-making and prediction.
   - V2X communication (V2V, V2I, V2P, V2N).
   - High-definition maps and GPS.
   - Cloud computing and OTA update systems.

9. What is Vehicle-to-Vehicle (V2V) communication? 5 points

   Ans: Direct wireless exchange of data between nearby vehicles. It shares speed, position, direction, and braking status. It helps prevent collisions and coordinate maneuvers. It also operates via DSRC or C-V2X protocols. Enhances situational awareness beyond line-of-sight.

10. What is Vehicle-to-Infrastructure (V2I) communication? 5 points

Ans: Communication between vehicles and road infrastructure (traffic lights, signs, cameras). It enables smart traffic signal timing and congestion alerts. ANd supports emergency vehicle preemption and road hazard warnings. And improves urban mobility and reduces emissions.

11. How can Intelligent Connected Vehicles help reduce road accidents? 5 points

    Ans: It predicts and warns of potential collisions via sensor fusion. And it automates braking or steering in emergencies. Share real-time hazard data with nearby vehicles. Reduce human error through automation. Improve response to sudden road conditions (e.g., black ice, accidents).

12. What are some applications of Intelligent Connected Vehicles in daily life? 5 points

    Ans:

    - Smart parking guidance and reservation.
    - Real-time traffic avoidance and route optimization.
    - Automated toll payments and congestion charging.
    - Emergency response coordination (e.g., automatic crash alerts).
    - Personalized infotainment and in-car services.

13. Why is Artificial Intelligence important in the development of ICVs? 5 points

    Ans: Because it enables real-time decision-making from sensor data. And learns from driving patterns and improves over time. And also predicts pedestrian/vehicle behavior for safer navigation. Manages complex scenarios (e.g., merging, roundabouts). Powers autonomous driving systems and voice assistants.

14. What happens during the Fully Autonomous and Intelligent Phase? 5 points

    Ans: No human driver required — vehicle operates entirely on its own. Handles all driving tasks in all conditions. Integrates AI, V2X, and HD maps for seamless operation. Can be summoned remotely or used as a robo-taxi. Redefines car ownership and urban mobility.

15. How can ICVs improve traffic management in smart cities? 5 points

    Ans: It optimizes traffic light timing based on real-time flow. And reduce congestion via coordinated vehicle routing. Also enable dynamic lane management and incident response. And Lower emissions through smoother traffic flow. Can integrate with public transit and mobility-as-a-service platforms.